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== Vitis HLS Report for 'matVecMul'
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* Date: Mon May 4 03:06:58 2026

* Version: 2021.1 (Build 3247384 on Thu Jun 10 19:36:07 MDT 2021)
* Project: matVecMul_stream_prj
* Solution: solution3 (Vivado IP Flow Target)
* Product family: zynq
* Target device: xc7z020-clg400-1

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== Performance Estimates
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+ Timing:
 * Summary:

Clock	Target	Estimated	Uncertainty
ap_clk	10.00 ns	5.831 ns	2.70 ns

+ Latency:
 * Summary:

Latency (cycles) min	Latency (cycles) max	Latency (absolute) min	Latency (absolute) max	Interval min	Interval max	Pipeline Type
20742	20742	0.207 ms	0.207 ms	20743	20743	no

 + Detail:
 * Instance:

Instance	Module	Latency (cycles) min	Latency (cycles) max	Latency (absolute) min	Latency (absolute) max	Interval min	Interval max	Pipeline Type
grp_matVecMul_Pipeline_sample_loop_fu_261	matVecMul_Pipeline_sample_loop	66	66	0.660 us	0.660 us	66	66	no

 * Loop:

Loop Name	Latency (cycles) min	Latency (cycles) max	Iteration Latency	Initiation Interval achieved	Initiation Interval target	Trip Count	Pipelined
- matmul_outer_loop	20672	20672	323	-	-	64	no
+ matmul_inner_loop	320	320	5	-	-	64	no

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== Utilization Estimates
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* Summary:

Name	BRAM_18K	DSP	FF	LUT	URAM
DSP	-	1	-	-	-
Expression	-	-	0	460	-
FIFO	-	-	-	-	-
Instance	4	-	201	310	-
Memory	1	-	0	0	0
Multiplexer	-	-	-	147	-

Register	-	-	230	-	-
Total	5	1	431	917	0
Available	280	220	106400	53200	0
Utilization (%)	1	~0	~0	1	0

+ Detail:

* Instance:

Instance	Module	BRAM_18K	DSP	FF	LUT	URAM
control_s_axi_U	control_s_axi	4	0	192	238	0
grp_matVecMul_Pipeline_sample_loop_fu_261	matVecMul_Pipeline_sample_loop	0	0	9	72	0
Total		4	0	201	310	0

* DSP:

Instance	Module	Expression
mul_mul_16s_16s_32_4_1_U10	mul_mul_16s_16s_32_4_1	i0 * i1

* Memory:

Memory	Module	BRAM_18K	FF	LUT	URAM	Words	Bits	Banks	W*Bits*Banks
x_localbuff_V_U	x_localbuff_V	1	0	0	0	64	16	1	1024
Total		1	0	0	0	64	16	1	1024

* FIFO:
N/A

* Expression:

Variable Name	Operation	DSP	FF	LUT	Bitwidth P0	Bitwidth P1
acc_V_4_fu_642_p2	+	0	0	45	38	38
add_ln1169_fu_350_p2	+	0	0	14	13	13
add_ln34_fu_300_p2	+	0	0	39	32	1
add_ln39_fu_332_p2	+	0	0	14	7	1
p_Val2_1_fu_420_p2	+	0	0	23	16	16
ret_V_fu_628_p2	+	0	0	46	39	39
sub_fu_287_p2	+	0	0	39	32	2
and_ln789_fu_506_p2	and	0	0	2	1	1
and_ln790_fu_520_p2	and	0	0	2	1	1
and_ln795_fu_550_p2	and	0	0	2	1	1
ap_block_state6	and	0	0	2	1	1
carry_1_fu_440_p2	and	0	0	2	1	1
overflow_1_fu_544_p2	and	0	0	2	1	1
overflow_fu_662_p2	and	0	0	2	1	1
underflow_fu_568_p2	and	0	0	2	1	1
Range1_all_ones_fu_472_p2	icmp	0	0	12	14	2
Range1_all_zeros_fu_478_p2	icmp	0	0	12	14	1
Range2_all_ones_fu_456_p2	icmp	0	0	12	13	2
icmp_ln34_fu_295_p2	icmp	0	0	18	32	32
icmp_ln39_fu_327_p2	icmp	0	0	18	32	32
out_stream_TLAST_int_regslice	icmp	0	0	18	32	32
or_ln384_fu_582_p2	or	0	0	2	1	1
or_ln794_fu_532_p2	or	0	0	2	1	1
or_ln795_fu_556_p2	or	0	0	2	1	1

acc_V_fu_682_p3	select	0	0	38	1	38
deleted_ones_fu_512_p3	select	0	0	2	1	1
deleted_zeros_fu_484_p3	select	0	0	2	1	1
select_ln384_1_fu_574_p3	select	0	0	17	1	15
select_ln384_fu_674_p3	select	0	0	39	1	37
val_V_fu_588_p3	select	0	0	16	1	16
xor_ln340_fu_668_p2	xor	0	0	2	1	1
xor_ln416_fu_434_p2	xor	0	0	2	1	2
xor_ln789_fu_500_p2	xor	0	0	2	1	2
xor_ln794_1_fu_526_p2	xor	0	0	2	1	2
xor_ln794_2_fu_538_p2	xor	0	0	2	1	2
xor_ln794_fu_656_p2	xor	0	0	2	1	2
xor_ln795_fu_562_p2	xor	0	0	2	1	2
Total		0	0	460	338	343

* Multiplexer:

Name	LUT	Input Size	Bits	Total Bits
acc_V_3_reg_249	9	2	38	76
ap_NS_fsm	65	12	1	12
i_1_fu_166	9	2	32	64
in_stream_TREADY_int_regslice	9	2	1	2
j_reg_238	9	2	7	14
out_stream_TDATA_blk_n	9	2	1	2
x_localbuff_V_address0	14	3	6	18
x_localbuff_V_ce0	14	3	1	3
x_localbuff_V_we0	9	2	1	2
Total	147	30	88	193

* Register:

Name	FF	LUT	Bits	Const Bits
acc_V_3_reg_249	38	0	38	0
add_ln34_reg_728	32	0	32	0
add_ln39_reg_741	7	0	7	0
ap_CS_fsm	11	0	11	0
col_size_read_reg_705	32	0	32	0
grp_matVecMul_Pipeline_sample_loop_fu_261_ap_start_reg	1	0	1	0
i_1_fu_166	32	0	32	0
j_reg_238	7	0	7	0
row_size_read_reg_711	32	0	32	0
shl_ln_reg_733	6	0	13	7
sub_reg_717	32	0	32	0
Total	230	0	237	7

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== Interface
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* Summary:

RTL Ports	Dir	Bits	Protocol	Source Object	C Type
s_axi_control_AWVALID	in	1	s_axi	control	array
s_axi_control_AWREADY	out	1	s_axi	control	array
s_axi_control_AWADDR	in	14	s_axi	control	array
s_axi_control_WVALID	in	1	s_axi	control	array
s_axi_control_WREADY	out	1	s_axi	control	array

s_axi_control_WDATA	in	32	s_axi	control	array
s_axi_control_WSTRB	in	4	s_axi	control	array
s_axi_control_ARVALID	in	1	s_axi	control	array
s_axi_control_ARREADY	out	1	s_axi	control	array
s_axi_control_ARADDR	in	14	s_axi	control	array
s_axi_control_RVALID	out	1	s_axi	control	array
s_axi_control_RREADY	in	1	s_axi	control	array
s_axi_control_RDATA	out	32	s_axi	control	array
s_axi_control_RRESP	out	2	s_axi	control	array
s_axi_control_BVALID	out	1	s_axi	control	array
s_axi_control_BREADY	in	1	s_axi	control	array
s_axi_control_BRESP	out	2	s_axi	control	array
ap_clk	in	1	ap_ctrl_hs	matVecMul	return value
ap_rst_n	in	1	ap_ctrl_hs	matVecMul	return value
interrupt	out	1	ap_ctrl_hs	matVecMul	return value
in_stream_TDATA	in	32	axis	in_stream_V_data_V	pointer
in_stream_TVALID	in	1	axis	in_stream_V_dest_V	pointer
in_stream_TREADY	out	1	axis	in_stream_V_dest_V	pointer
in_stream_TDEST	in	1	axis	in_stream_V_dest_V	pointer
in_stream_TKEEP	in	4	axis	in_stream_V_keep_V	pointer
in_stream_TSTRB	in	4	axis	in_stream_V_strb_V	pointer
in_stream_TUSER	in	1	axis	in_stream_V_user_V	pointer
in_stream_TLAST	in	1	axis	in_stream_V_last_V	pointer
in_stream_TID	in	1	axis	in_stream_V_id_V	pointer
out_stream_TDATA	out	32	axis	out_stream_V_data_V	pointer
out_stream_TVALID	out	1	axis	out_stream_V_dest_V	pointer
out_stream_TREADY	in	1	axis	out_stream_V_dest_V	pointer
out_stream_TDEST	out	1	axis	out_stream_V_dest_V	pointer
out_stream_TKEEP	out	4	axis	out_stream_V_keep_V	pointer
out_stream_TSTRB	out	4	axis	out_stream_V_strb_V	pointer
out_stream_TUSER	out	1	axis	out_stream_V_user_V	pointer
out_stream_TLAST	out	1	axis	out_stream_V_last_V	pointer
out_stream_TID	out	1	axis	out_stream_V_id_V	pointer